

Markscheme

Mathematics: analysis and approaches

Practice question bank

19. (a)

Consider the function $f(x) = \frac{5x - 2}{(x - 4)(x + 5)}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x - 4} + \frac{B}{x + 5}$.

$$5x - 2 = A(x + 5) + B(x - 4) \quad (M1)$$

$$\text{substituting } x = 4: 18 = 9A \Rightarrow A = 2 \quad \mathbf{A1}$$

$$\text{substituting } x = -5: -27 = -9B \Rightarrow B = 3 \quad \mathbf{A1}$$

[3 marks]